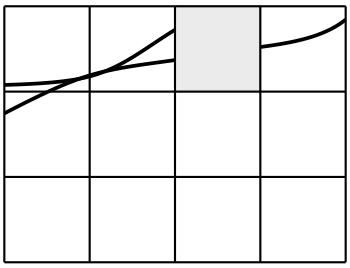
Preview: set-03/by_level/Grade_03-04/y2002_q01 (v3)

Find the missing square by its edges: lines must continue at the same places, not turned

Part A: Each question is worth 3 points

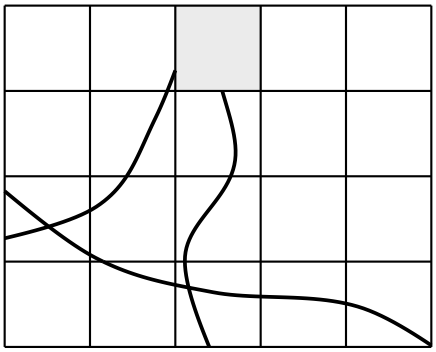
1. A cat played with balls of wool on squared paper. Now one square of the picture is missing.



Which square was cut out of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

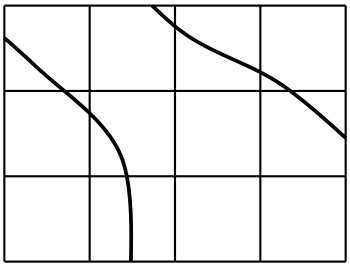
2. Mia drew some lines on squared paper. Then she cut out one square of her picture.



Which square was cut out of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

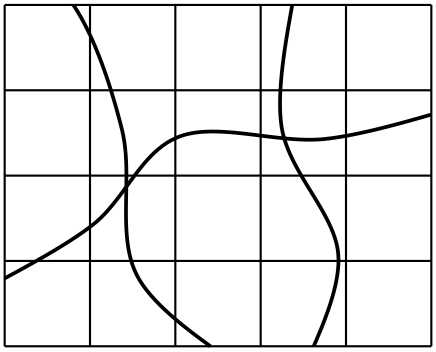
3. A cat played with balls of wool on squared paper. Here is the whole picture.



Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

4. Leo painted a river and some roads on a map made of squares. Here is the whole picture.

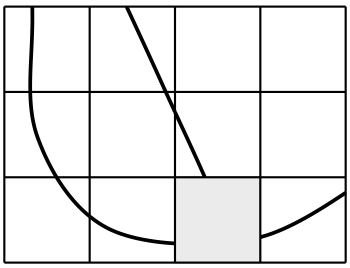


Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part B: Each question is worth 4 points

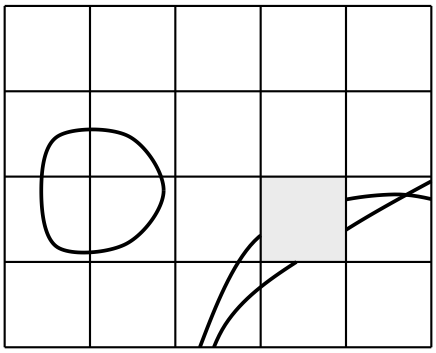
5. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).



Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

6. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).

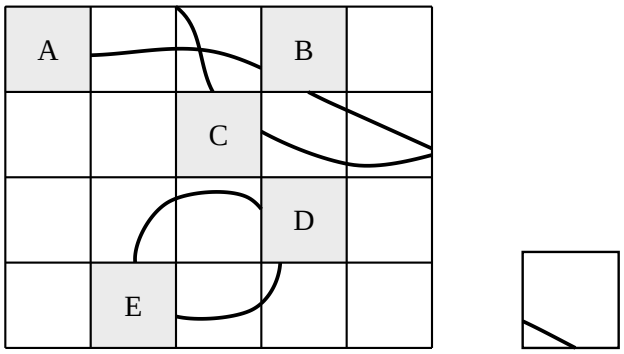


Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part C: Each question is worth 5 points

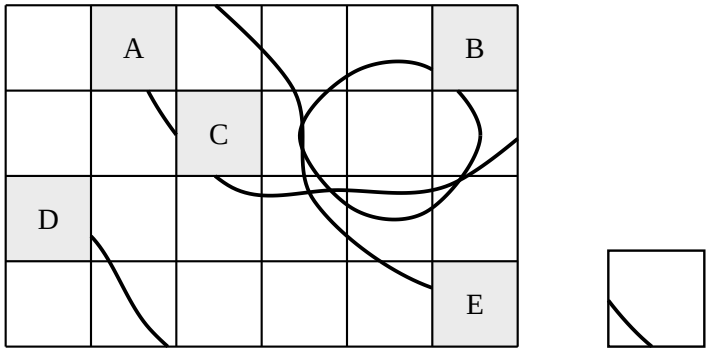
7. Leo painted a river and some roads on a map made of squares. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

8. Some paths go across a park. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

Answer Key

1. **D**	**2. A**	**3. E**	**4. A**	**5. E**	**6. E**
7. **B**	**8. C**				

Solutions

1. D — Look where the lines of the picture reach the grey gap: top: 1 line, right: 1 line, bottom: no line, left: 2 lines. Only square D has its lines in exactly those places without being turned. The others don't fit: 3 are from another place, one is mirrored. [set-03/by_level/Grade_03-04/2002 Q1 · missing_square]

2. A — Look where the lines of the picture reach the grey gap: top: 2 lines, right: no line, bottom: 1 line, left: 1 line. Only square A has its lines in exactly those places without being turned. The others don't fit: 2 are from another place, one is mirrored and turned, one is upside down. [set-03/by_level/Grade_03-04/2002 Q1 · missing_square]

3. E — Find each square in the picture by the places where its lines reach the edges. Squares A, B, C, D can be found; square E is a piece turned, so it is not in the picture like that. [set-03/by_level/Grade_03-04/2002 Q1 · not_a_piece]

4. A — Find each square in the picture by the places where its lines reach the edges. Squares B, C, D, E can be found; square A is a piece mirrored and turned, so it is not in the picture like that. [set-03/by_level/Grade_03-04/2002 Q1 · not_a_piece]

5. E — Look where the lines of the picture reach the grey gap: top: 1 line, right: 1 line, bottom: 1 line, left: 1 line. Only square E has its lines in exactly those places in this position. The others don't fit: one is mirrored and turned, 2 are turned, one is upside down. [set-03/by_level/Grade_03-04/2002 Q1 · turned_square]

6. E — Look where the lines of the picture reach the grey gap: top: no line, right: 2 lines, bottom: 1 line, left: 1 line. Only square E has its lines in exactly those places in this position. The others don't fit: one is mirrored, one is mirrored and turned, one is mirrored and upside down, one is turned. [set-03/by_level/Grade_03-04/2002 Q1 · turned_square]

7. B (gap B) — The square has top: no line, right: no line, bottom: 1 line, left: 1 line. Only gap B has lines reaching it at exactly those places. [set-03/by_level/Grade_03-04/2002 Q1 · which_gap]

8. C (gap C) — The square has top: no line, right: no line, bottom: 1 line, left: 1 line. Only gap C has lines reaching it at exactly those places. [set-03/by_level/Grade_03-04/2002 Q1 · which_gap]